

Privacy-Preserving Technologies for Diversity Analytics

Document type	Technical architecture guide
Coverage	Cultural Infusion Atlas - synthetic research snapshot
Topics	differential privacy homomorphic encryption federated learning SMPC anonymisation
Source inspiration	https://atlas.culturalinfusion.com/secure-diversity-inclusion-survey-trust/

This document is a synthetic test artifact derived from themes and publicly visible metadata in the supplied Atlas article. It contains paraphrase, explanatory expansion and invented scenarios. It is not legal advice, a real client record, or a substitute for the original source and linked academic paper.

Threat model

A privacy architecture begins with plausible threats: an external attacker steals a database; an administrator exports raw responses; a manager isolates a small team through filters; a colleague recognises a free-text event; or repeated aggregate queries reveal an individual's contribution. Encryption alone addresses only part of this set.

The protected assets include identity attributes, survey answers, authentication records, organisation structure, free text, report exports, query history and cryptographic keys. Adversaries can possess auxiliary knowledge such as a person's team, location or public identity. Controls should be assessed against those combinations.

Data minimisation and pseudonymisation

Minimisation removes fields that do not support a documented objective and limits precision where exact values are unnecessary. Pseudonymisation replaces direct identifiers with controlled references so operational workflows can be separated from analysis. It reduces exposure but remains personal-data processing when re-linking is possible.

A synthetic survey stores invitation delivery in one service and responses in another. The analysis dataset uses a rotating participant token. The token mapping is deleted after reminder periods close. This design reduces routine linkage but does not claim irreversible anonymisation while contextual attributes remain.

Aggregation and small-cell suppression

Aggregation restricts reports to groups rather than individuals. Small-cell suppression hides results below a defined threshold and may also suppress complementary cells so totals do not reveal the missing value. Query controls must account for differencing attacks in which two overlapping reports isolate a person.

The fictional Atlas dashboard starts with a threshold of ten for general results and fifteen for high-sensitivity cross-tabs. An audit finds that repeated filters can still infer a group of two, so the revised design tracks

query sets and applies a contribution rule. Thresholds are examples for testing, not production recommendations.

Differential privacy

Differential privacy bounds how much a published result can change when one person's record is added or removed. A mechanism adds calibrated randomness according to sensitivity and a privacy parameter commonly called epsilon. Smaller epsilon generally means stronger privacy and lower accuracy.

A privacy budget limits cumulative disclosure across repeated releases. Product teams must decide which queries consume budget, how to communicate uncertainty and what happens when the budget is exhausted. Differential privacy is not a magic anonymisation label; poor clipping, unlimited exports or leakage through non-private features can invalidate the protection.

Homomorphic encryption and secure multi-party computation

Homomorphic encryption supports selected computations over encrypted values without exposing plaintext to the computing service. Secure multi-party computation lets parties jointly compute an output while keeping their inputs separate. Both can reduce trust placed in a central processor but introduce performance, implementation and key-management complexity.

A fictional consortium uses secure aggregation to combine inclusion indicators across member organisations without sharing employee-level records. The protocol reveals only totals above a publication threshold. Governance still matters because a malicious participant can submit manipulated inputs or infer information when the consortium is too small.

Federated learning

Federated learning trains a shared model across separated data stores. Participants exchange model updates rather than raw records. Updates can nevertheless leak information, so secure aggregation, clipping, differential privacy and participant authentication may be required.

For many diversity-survey tasks, ordinary aggregated statistics may be more transparent and proportionate than federated machine learning. The architecture should choose the simplest approach that meets the analytical goal and threat model.

Encryption and key management

Transport encryption protects network paths, while storage encryption protects media and backups. Application-layer encryption can reduce exposure to infrastructure administrators. Key rotation, separation of duties, recovery testing, hardware-backed storage and access logging are essential supporting controls.

Encryption does not decide whether an analyst should see a record. Authorisation, report design, export controls and monitoring remain necessary. A platform should be tested for indirect paths such as logs, caches, support tools and analytics telemetry.

Technology selection record

The synthetic decision record selects minimisation, pseudonymisation, role-based access, aggregation thresholds, monitored query controls and encryption as baseline measures. Differential privacy is reserved for external benchmark releases. Homomorphic encryption and federated learning are evaluated but not adopted because the use case does not justify their operational complexity.

Bob should retrieve this decision when asked which advanced techniques were rejected, while separately explaining that the source article lists them as emerging options. That distinction tests whether generation respects document context instead of combining every retrieved technology into one recommendation.